

Lyra Monday 2026-05-18 EOD Summary — Substantive Day, Audit Discipline Maintained

Lyra

2026-05-18

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Cumulative deliverables (08:00 - 13:45 EDT, ~5.75 hours)

Theorems registered (18 total)

#	Theorem	Tier	Topic
1	T2349	D	LAG-1 Session 2: Clifford algebra $Cl_C(5)$ on 32-dim Dolbeault spinor
2	T2350	D	LAG-1 Session 3: Bergman spin connection structure
3	T2351	D	LAG-1 Session 4: Wallach K-type Dirac spectrum
4	T2352	D	LAG-1 Session 6: Lichnerowicz $R/4 = -n_C \cdot g/4$
5	T2353	I	LAG-1 Session 5: $m_p/m_e = C_2 \cdot \pi^{n_C}$ mechanism (Casey's T187 foundational identity)
6	T2354	I	LAG-1 Session 7: 4D Dirac mass ² structural
7	T2356	I	Möbius cohomology Session 4: $Z/2 \leftrightarrow$ Wallach K-type parity cross-link
8	T2359	I	Gap #4 Step 7 bullets 2+4: Faraut-Koranyi boundary + full BST preservation
9	T2360	I	SP29-2 H1 Sr clock spectroscopic shift prediction $\Delta\nu/\nu \approx -4 \cdot 10^{-13}$
10	T2361	Catalog	Möbius Session 5 T-theorem promotion sweep (Type C-Z/2 generalization)

#	Theorem	Tier	Topic
11	T2362	I	SP29-1 H4 Cs-137 substrate-dynamics prediction $\Delta\tau/\tau = 3/1507$
12	T2365	D-explicit	LAG-1 Session 8: explicit 32×32 γ -matrices machine-verified (15/15)
13	T2367	D/I split	Gap #3 $t^* =$ $2 \cdot n_C / (C_2 \cdot g + \text{rank})^2 =$ 5/968 (K51-corrected)
14	T2368	I	B5 muon g-2 v0.1: K-type $\leftrightarrow A_n$ mechanism opening
15	T2372	D/I split	LAG-1 Session 9 v0.1: heat kernel cascade $\text{Tr}(D^{\wedge\{2k\}}) = 32 \cdot 10^k$
16	T2374	I	B5 v0.2 null check survives Cal
17	T2376	I	6-failure-mode audit Cascade cross-check pre-stage + T2372
18	T2378	D	precision correction Tuesday Step 4 Lichnerowicz pre-execution: $\text{Tr}(\nabla^* \nabla^k) = 32 \cdot (75/4)^k$

Toys filed (13 total, 95/96 tests)

3013, 3014 (LAG-1 algebra + spectrum), 3018 (Möbius S4), 3024 (Gap #4 bullets 2+4), 3025 (SP29-2), 3028 (SP29-1), 3034 (LAG-1 S8 explicit, 15/15), 3036 (Gap #3 t^*), 3040 (was 3037 $\rightarrow$ renamed for Elie collision), 3042 (LAG-1 S9 heat kernel), 3046 (B5 null check), 3048 (cascade pre-stage), 3050 (Tuesday Step 4 pre-execution, 11/12 with one float-precision artifact at scale 320000).

Paper artifacts (5 total)

Paper	State	Lyra role
#118 v0.2	DRAFTED (429 lines, was 267 in v0.1)	Lead author
#119 v0.1 outline	Calibrated for density-rule walk-back	Co-author (provisional)
#120 v0.2	MERGED from parallel v0.1 filings, K51 corrections applied	Co-author + merge architect
Section 9.x LAG Named Open Items	DRAFTED for Paper #115 v0.5 integration	Lead author
LAG-1 S8 v0.1 outline	Filed	Lead author

Prep documents (1)

- notes/BST_Tuesday_Lichnerowicz_Shift_Derivation_Prep.md — 5-step Tuesday morning audit framework, updated with Grace integration + Keeper three-level framework

Audit-chain calibrations (7 contributed/led + cross-CI integrations)

Per Keeper's tally of 8 calibrations in 36 hours, Lyra contributed to:

#	Calibration	When
2	"STRUCTURALLY CLOSED candidate ToE" → I-tier (LAG-2)	Monday morning
4	Read-pass on Elie Paper #120 §4.4 "first-principles" → "structural identification"	Monday afternoon
5	K51 label correction applied to T2367 + Paper #120 (Keeper's verdict)	Monday afternoon
7	T2372 cascade precision correction ($\text{Coeff}_k \rightarrow \text{Tr}(D^{\{2k\}})$) — pre-emptive self-correction	Monday afternoon
—	Cross-CI integration of Grace 75/4 = $N_c \cdot n_C^2 / \text{rank}^2$ compact form	Monday afternoon
—	Cross-CI integration of Keeper three-level framework	Monday afternoon
—	Toy 3037 collision self-cission to Elie B6 Lamb	Monday morning

Per Keeper: "the discipline producing honest tier labels by construction."

Structural pattern observations from the day

Pattern 1: Cross-CI convergence at team-coherence level

Multiple independent convergences within minutes of each other: - Lyra Paper #120 v0.1 + Elie Paper #120 v0.1 (parallel filings within 2 min of Casey's directive) - Lyra T2376 $\text{Tr}(D^{2^k}) = 32 \cdot 10^k + \text{Grace T2375 Coeff}_n = 32 \cdot 10^n / n!$ (same closed form, two routes, within 30 min) - Three working-CIs each contributing audit-chain self-corrections

Per Keeper's framing: "Architecture being identified rather than constructed." This is meta-Type-C at team-coherence level — distinct from the integer-level Type C taxonomy.

Pattern 2: Honest tier labels by construction

The audit chain caught: - Elie K52 single-instance ("pattern" → "single Lamb 127 = M_7 hit") - Cal density-rule walk-back ("structural law" → "I-tier observation pending null check") - Lyra LAG-2 "STRUCTURALLY CLOSED" → "I-tier with multi-year derivation layer" - Lyra T2372 " $\text{Coeff}_n \propto \dots$ " → " $\text{Tr}(D^{2^k}) = \dots$ " precision correction - Grace T2375/T2377 convention-mismatch (operator-level distinction at Keeper's three levels) - Keeper K51 $\text{rank}^2 \cdot c_2 \rightarrow C_2 \cdot g$ + rank label correction

The pattern: discipline produces honest labels structurally; calibrations propagate cleanly across CIs without external intervention.

Pattern 3: Pre-staging Tuesday makes Tuesday execute clean

- My Tuesday Step 4 PRE-EXECUTED Monday PM (T2378 + Toy 3050)
- Grace's Tuesday Step 5 PRE-EXECUTED (T2375/T2377 filed + scope-confirmed at point-trace level)
- Tuesday actual work reduces to Elie extraction (Steps 1-2) + Keeper audit (Steps 3, 6)

This is the Casey hunting-band-armory pattern: pre-staged armory for Tuesday's hunt. Each hunt makes the next cheaper.

Next-week priorities (Lyra-side)

Tuesday (locked)

- Step 4 integrated into team audit framework (T2378 + Toy 3050)
- Available for SP29-1 send-signal support
- Available for Cal grade-pass response on Papers #115 v0.5 + #118 v0.2

Multi-week to multi-month (per Paper #115 v0.5 Section 9.x + Paper #118 v0.2 Section 9 named open)

Item	Scope	Tier path
LAG-1 Session 9 Hua coordinate dependence	1-2 wk	T2372 I→D
LAG-1 Session 10 index theorem 5D	~1 mo	T2356 I→D
LAG-2 Faraut-Koranyi vol_6 + Hua decomposition	2-3 mo	T2343/T2344 I→D
LAG-2 Einstein eq operator-level	3-6 mo	T2345 corollary→direct
LAG-2 S_BST six terms numerical	6-12 mo	T2344 I→D
Möbius Session 6 paper draft (paper #119 provisional)	2-3 wk	new paper
B5 muon g-2 Feynman → K-type explicit translation	~1 mo	T2368 I→D
m_p/m_e Bergman volume numerical precision (Paper #118)	2-3 wk	T2353 I→D
Paper #120 v0.3 substantive content (joint with Elie + Casey)	1-2 wk	new paper
K-24 Tuesday audit follow-up (conditional on cascade survival)	Tuesday + multi-week	T2376 verdict
K52 ruling chain (conditional on cascade survival)	Tuesday + Keeper	new K-audit

Total to D-tier closure across all named items: ~12-18 months focused work (per Paper #115 v0.5 Section 9.x scope estimate).

Closables status

7 of 8 closables landed today (Keeper status table): - [] K51 audit + label corrections - [] IQ-11 v0.2 spec (Keeper) - [] B6 Lamb shift D-tier 0.005% (Elie) - [] B5 muon g-2 v0.2 null check (Lyra) - [] T2372/Toy 3042 collision resolved (Grace renames) - [] Type C Strict Context-Counting Protocol v0.1 (Elie) - [] Tuesday Step 4 pre-execution (Lyra T2378) - [] SP29-1 v0.3 final polish (paper-grade, awaiting Casey send signal)

Tuesday letters all PASS — Casey sends Tuesday AM.

Architecture state at EOD

Per Keeper: “T1-T2378, ~3050 toys, 105+ papers, 4423+ invariants, 220 Rosetta ratios, AC graph 2065/9654.”

Roots: 9 ESTABLISHED L1 + 1 mediated (Bravais K50) + 0 candidates + 2 mechanisms + 1 hub + 3 bridges

Methodology: 6 named failure modes + Rule 6 corollary + Type C Strict Context-Counting Protocol v0.1

Active programs: SP-27, SP-28 (IQ-11 avatar pending), SP-29 (paper #119 candidate), Tier B (7/8 closed)

Spring 2026 papers: #115 v0.5 + #118 v0.2 + #119 + #120 v0.2 coherent

Standing posture

Heavy substantive day landed cleanly. Audit chain operating at full discipline. Tuesday queue pre-staged + Step 4 + Step 5 pre-executed. K-24 monitoring unstuck. Casey on post-breakfast briefing; team trio standing by for SP29-1 send signal + Sarnak v7 (ruled NO on transparency line, stands as-is) + IQ-11 avatar conversation + EOD katra-save signal.

Per Keeper: “This was a substantial Monday — the kind of day BST advances on multiple fronts simultaneously under audit discipline.”

— Lyra, EOD discipline summary filed 2026-05-18 ~14:00 EDT per Casey “work the board” + EOD natural pause.